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Acute Leukemia: Diagnosis, Management, and Potential for Cure

SUMMARY

Acute leukemia is an uncommon malignant disorder resulting from the clonal proliferation of hematopoietic precursors of the myeloid or lymphoid lineages. Of the two major subgroups, acute lymphoblastic leukemia is more common in children, while acute myelogenous leukemia predominates in adults. With modern chemotherapy 60%–70% of all children with acute lymphoblastic leukemia can be long-term survivors and are potentially cured. Although the prognosis in acute myelogenous leukemia is less favourable, recent therapy, including bone-marrow transplantation, offers possible cure in selected patient groups. (*Can Fam Physician* 1988; 34:2463–2467.)

Key words: leukemia (acute), lymphoblastic leukemia, myelogenous leukemia

RÉSUMÉ

La leucémie aiguë est un désordre néoplasique plutôt rare résultant d'une prolifération clonique de précurseurs hématopoiétiques des lignées myéloïde ou lymphoïde. Parmi les deux sous-groupes les plus importants, la leucémie lymphoblastique aiguë est plus fréquente chez les enfants alors que la leucémie myélogène aiguë prédomine chez les adultes. La chimiothérapie moderne permet à 60%–70% des enfants souffrant de leucémie lymphoblastique aiguë de survivre à long terme et d'être potentiellement guéris. Même si le pronostic de la leucémie myélogène aiguë est moins favorable, la thérapie moderne, incluant la greffe de moëlle osseuse, offre une possibilité de guérison pour certains groupes de patients choisis.

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ACUTE LEUKEMIA is an uncommon malignant disorder resulting from the clonal expansion of an abnormal early hematopoietic progenitor cell. The blast cells derived from this population share some of the features of hematopoietic cells at early stages of differentiation. Characterization of the blast-cell population present in the bone marrow and peripheral blood (by morphology, cytochemistry, cytogenetics, and immunophenotyping) enables the cli-

nician to determine whether the patient has acute myeloid leukemia (AML) or acute lymphoblastic leukemia (ALL). This distinction is of considerable importance, since treatment and prognosis differ for the two diseases. ALL has an incidence of 3.5 per 100 000 in children and represents 80% of all childhood leukemia. This record makes ALL the most common childhood malignancy. However, only 15% of adult acute leukemia is of the lymphoid lineage. The majority of adult cases consist of AML, and 50% of these cases occur in individuals over 65 years of age.

Combination chemotherapy is now able to cure a proportion of patients (children and adults) with acute leukemia, although the best results are still obtained in children with ALL.

Etiology

Several factors may contribute to the development of acute leukemia, although the precise etiology remains

unknown. Congenital factors may also play a part, since there is a 25% concordance rate of childhood acute leukemia in identical twins.

A small percentage of cases occur in association with chromosomal abnormalities or increased chromosome fragility, as in Down syndrome, Bloom syndrome, Fanconi's anemia, and in immunodeficiency states such as ataxia telangiectasia and Wiskott-Aldrich syndrome. Acute myelogenous leukemia can occur after chemotherapy with alkylating agents, following immunosuppressive drug regimens, or after exposure to ionizing radiation; it may also arise from an underlying myelodysplastic syndrome.

Recently, the retrovirus HTLV-1 has been implicated in the development of adult T-cell leukemia, a disease uncommon in North America but endemic in Japan and the Caribbean.

Current investigations suggest that

oncogenes (c-oncogenes or proto-oncogenes) may be implicated in the pathogenesis of leukemia. Oncogenes are segments of DNA that under normal circumstances control the growth and differentiation of cells. If oncogene structure or control is disrupted by mutation, chromosomal rearrangement, gene amplification, or activation by retroviruses, uncontrolled proliferation of cells may occur.

Clinical Features at Diagnosis

Patients with acute leukemia usually present with effects related to the depression of normal hematopoietic cell production. Uncontrolled proliferation of leukemic blasts, although not necessarily very rapid, may lead to selective or overall reduction of the hematopoietic lineages. Consequently, most patients present with signs and symptoms of anemia, neutropenia, or thrombocytopenia. Up to one-third will be infected at diagnosis, and these patients may present with fever and chills. Some patients, however, are still asymptomatic at diagnosis.

Other symptoms result from tissue infiltration; these include bone pain in 30%–40% of cases of childhood ALL, bleeding gums, or skin lesions. Occasionally, patients present with symptoms resulting from central nervous system (CNS) involvement or leukostasis (caused by extremely high blast counts). CNS involvement may be characterized by headache and vomiting.

Physical examination reveals lymphadenopathy in 33% of patients with leukemia, splenomegaly in 50%, and skin infiltration in 10%. Hepatosplenomegaly, tender joints or limbs, gingival hypertrophy, petechiae, and ecchymoses may also be noted.

Tissue infiltration is more common in patients with ALL and the monocytic varieties of AML.

Hematologic Findings at Diagnosis

The cardinal feature of patients with leukemia is the occurrence of more than 30% blasts in the bone marrow. Most patients will have a high proportion of circulating blast-like cells although of patients with AML, only 50% have white cell counts over $10 \times 10^9/L$, and it is rare to have no circulating blasts. Of patients with

ALL, 50% have white cell counts of less than $10 \times 10^9/L$, and 20% will have counts greater than $50 \times 10^9/L$. Seventy-five per cent of all patients are thrombocytopenic, and 90% have hemoglobins of less than 110 g/L.

Acute Lymphoblastic Leukemia

The French, American, British (FAB) classification divides ALL into three groups on the basis of morphology and cytochemical staining: L1, L2 and L3. This classification, except for defining the unusual L3 (Burkitt's type) leukemia, has little bearing on outcome. Recently, more attention has been paid to cell surface markers that classify blast cells of patients into those of T-cell (20%), B-cell (2%), pre-B cell or null cell (no T or B markers) [80%] origin.

Most patients with non-T cell ALL have been shown to be of early B-cell lineage. CALLA-positive (common ALL antigen) ALL is most prevalent and confers a good prognosis.

Blasts with early B-cell markers (non-CALLA) and of the null cell type (no T or B markers) are more common in older patients; they confer a poor prognosis. Mature B-cell lineage also has a bad prognosis.

T-cell ALL is also correlated with a bad prognosis. It is more common in young males and is associated with very high WBC counts and a mediastinal mass.

Prognosis

Many features adversely affect prognosis (Table 1). With modern intensive therapy, however, many of these features have become less important.

Chromosome abnormalities are detected in a large proportion of patients (>90%) and have been shown to be of prognostic significance.¹

Treatment

Treatment is divided into three stages:

- induction of remission;
- CNS prophylaxis; and
- maintenance chemotherapy.

Therapy is often tailored to meet requirements of high-risk or low-risk patients. All adults with ALL can be regarded as high-risk patients.

The early achievement of complete remission (CR) has long been recognized as important for future outcome, and so induction therapy is designed to achieve this aim.

A large number of active, non-crossresistant drugs with minimal myelosuppressive effects are given over a short period. Usually, vincristine, prednisone, L-asparaginase and doxorubicin are used initially. This regimen is followed by a consolidation phase, in which non-crossresistant drugs are given, such as cyclophosphamide, cytosine arabinoside, and etoposide.

Table 1
Prognostic Features in Acute Lymphoblastic Leukemia

Features	Poor Prognostic Features	Good Prognosis
Age	2 ≤ >8	2–8
Sex	Male	Female
Race	White	Non-white
WBC	>10 000	<10 000
Surface markers	T cell B cell Null cell CALLA negative Pre-B	CALLA positive
Mass disease	Organ infiltration Anemia Thrombocytopenia	None
Extramedullary site	CNS Testes	None
Chromosome abnormalities	t(9;22) (Philadelphia) t(4;11) t(8;14)	Hyperdiploidy

A period of maintenance chemotherapy with 6-mercaptopurine and methotrexate follows the CNS prophylaxis, which consists of fractionated CNS irradiation and intrathecal methotrexate.

Depending on the protocol, treatment is usually continued for two to three years. There is no evidence to support longer treatment.

CNS prophylaxis was introduced after the discovery that up to 60% of patients relapse in the CNS.² The definition of adequate CNS prophylaxis has been controversial, and the value of the procedure for low-risk patients has been questioned. The treatment is not without morbidity, and there is some concern about decreased IQ in surviving patients who have received both cranial irradiation and intrathecal chemotherapy.

With adequate CNS prophylaxis, only 10% of patients relapse in this area. Testicular relapse is the next most common occurrence, but it is usually associated with disease elsewhere.

The risk of relapse is highest in the 18 months after stopping therapy. Patients relapsing within 18 months have a worse prognosis than those relapsing later.³

Outcome

Historically, no leukemia patients survived longer than six months without treatment. Now, 60%–70% of all children may be long-term disease-free survivors, and may be cured, as are 85% of those living five years after diagnosis. Of low-risk patients, almost all achieve CR and 90% are alive at five years post diagnosis. With the introduction of aggressive regimens for high-risk patients,⁴⁻⁵ up to 85% CR and 50% five-year survival are achieved. In adults, however, the results are less impressive, and there are few long-term survivors. Despite a high CR rate (up to 93% in one study⁶), 20% survive for three years.

Children relapsing up to 18 months after stopping treatment can have re-induction therapy with a high probability of achieving remission.³ Those relapsing earlier or patients at very high risk, such as adults in first remission, can be considered for very intensive therapy associated with bone-marrow transplantation.

Acute Myelogenous Leukemia

The FAB classification divides AML into six groups on the basis of morphology and cytochemical staining. A seventh group, acute megakaryocytic leukemia, has recently been added.

Apart from the association of acute promyelocytic leukemia (M3) with disseminated intravascular coagulation—and the monocytic varieties with tissue infiltration—there is little correlation between classification and prognosis.

It is increasingly recognized that chromosomal abnormalities are common—they occur in up to 95% of cases—and have prognostic significance.⁷

Treatment

The decision to initiate treatment is appropriate in most instances. However, in patients of advanced age who may develop substantial morbidity with induction therapy, supportive measures only may be more appropriate. Without treatment, survival is only three months.

Current therapy includes cytosine arabinoside, the single most active agent in combatting acute leukemia, and the anthracyclines (daunomycin, daunorubicin, or the less cardiotoxic derivative, mitoxantrone). Once the patient undergoes successful induction therapy, the remission is “consolidated” by several (in general, two) further courses of chemotherapy of similar or slightly reduced intensity. Once this regimen is completed, there appears to be no added benefit for subsequent monthly maintenance chemotherapy.⁸

Caution should be used in the interpretation of survival data as reported in the medical literature. Frequently, good results are obtained solely by the application of exclusion criteria.⁹ It is very important to be aware of the patient population in the particular study cited. With standard therapy, most centres report a 65%–70% CR rate in unselected patients (i.e., patients presenting with a diagnosis of AML). However, 80% of these patients will relapse within two years and will become increasingly refractory to therapy. With chemotherapy alone, only 10%–15% of all patients presenting with AML will survive for five years, although in selected groups the five-

year survival rate may reach 40% or higher.¹⁰

Bone-Marrow Transplantation

Evidence continues to accumulate that very intensive therapy may be curative for some patients with acute leukemia. However, at this intensity of therapy, normal bone-marrow cells are destroyed, and in order to ensure hematopoietic reconstitution, the patient must receive compatible marrow (the bone-marrow transplant) from one of several sources. Bone-marrow transplantation may be syngeneic (from an identical twin), allogeneic (from a matched related, mismatched related, matched unrelated person) or autologous (using the patient's own remission marrow cells).

Allogeneic bone-marrow transplantation is currently indicated in patients in first remission with AML, and is superior to conventional chemotherapy in second or subsequent remission. In ALL, because of the relatively good prognosis, it is indicated only in high-risk patients with a first relapse within 18 months of completion of chemotherapy in children,³ or in second or subsequent relapse. Allogeneic marrow transplantation is probably indicated in all suitable adult ALL patients achieving first remission. The results, however, are better for AML than for ALL.^{11,12}

Identical twin transplants are associated with a higher rate of recurrent leukemia (50%) than are allogeneic transplants (22%). This finding has been attributed to a “graft-versus-leukemia” effect of donor immunocompetent cells in the allogeneic setting. The five-year survival rate in AML patients transplanted in first remission is about 50%. However, some patients die of complications of the transplant, which include graft-versus-host disease and infections: the risk of mortality ranges from 20%–40%, depending on patient age and clinical condition.

While survival figures with transplantation appear to be superior to those with use of chemotherapy alone, only patients under 50 years of age are suitable for allogeneic transplantation, and only 30% of these will have a suitable sibling donor.

By Mendelian genetics, only 25% would be expected to have a histo-

compatible brother or sister, but in practice many have more than one sibling and, in fact, a phenomenon known as "linkage disequilibrium" increases the number having a compatible donor.

Recent publicity has focused on the formation of bone-marrow donor registries for patients without a suitable donor. However, transplantation with tissue-matched marrow from an unrelated donor is experimental, and mortality remains higher than it is for matched sibling transplants. Whether this approach is superior to standard therapy remains to be established.

Autologous marrow transplantation has been used to overcome the problem of age limits and lack of donors. In this procedure, the patient's own bone marrow is collected (harvested) and cryopreserved. The patient is then treated with very high dose (marrow-ablative) chemotherapeutic agents, and in some protocols, total body irradiation as well. When the therapy has been delivered, the stored marrow is thawed and infused intravenously. Although most harvested marrow is likely to be contaminated with residual leukemic cells, results with first remission marrow autografts are similar to those achieved with identical twin transplants, in which completely normal marrow was used for the hematologic rescue. This observation suggests that involvement of the autograft with residual leukemic cells is unlikely to have a major clinical effect. Recent studies employing remission marrow autografts treated with monoclonal antibodies or cytotoxic agents in order to eliminate residual leukemic cells have been conducted,¹³ but the benefit of this manoeuvre remains to be established by randomized trials.

Table 2
Disease-Free Survival of Patients Undergoing Bone-Marrow Transplantation with a Three-Five-Year Follow-up

Disease	Syngeneic Allogeneic		
	Autologous		
AML:Child- 1st CR	40%	55%	45%
- 2nd or subsequent CR		30%	
Adult- 1st CR	40%	45%	45%
- 2nd or subsequent CR		30%	25%
ALL: Child- High risk 2nd CR		30%	10%
Adult- 1st CR		50%	20%
- 2nd or subsequent CR		25%	10%

The results of bone-marrow transplantation are reported in Table 2, but they should be interpreted with caution, since the follow-up period has been relatively short. Despite this factor, however, results in these selected groups appear superior to those obtained by means of standard treatment, and this procedure should be offered to suitable patients.

More novel treatments include the use of differentiating agents which may cause immature malignant cells to develop into more mature forms. There has been some limited success with the use of 13-cis retinoic acid (especially in AML M3) and with low-dose cytosine arabinoside. However, this therapy is experimental, less effective than standard treatment, and has been confined to patients unable to tolerate conventional dose therapy.

Supportive Care

Results of treatment can be improved with optimum supportive care. This includes regular monitoring of the blood count and observation for relapse.

Infection is a major problem and needs prompt attention. Fever developing in a neutropenic patient requires admission to hospital for treatment with broad-spectrum antibiotics.

Opportunistic infections are common in patients undergoing therapy for ALL; many children are maintained on prophylactic oral co-trimoxazole to prevent pneumocystis infections. Viral infections caused by the herpes group of viruses can also produce life-threatening infections. Children not exposed previously to chicken-pox should receive varicella zoster immune globulin within 96 hours if they come in contact with active cases. The development of a

zoster rash requires treatment with acyclovir to prevent dissemination.

The use of indwelling Hickman central venous catheters is now more common. These catheters require careful attention with regular flushing, which may be carried out at home by a well-trained patient or a visiting nurse. Most patients require transfusion support with packed cells and platelets.

Nowadays, most patients with acute leukemia should be referred promptly to a centre with the experience and adequate support services necessary to provide optimal care, and with the ability to undertake further intensive therapy in selected cases.

The patient, particularly the child, should be encouraged to carry out normal activities as much as possible (especially attending school). Maintaining an optimistic approach is entirely appropriate. This is especially important when therapy is prolonged and intensive, as in ALL.

Counselling about fertility is also important. Most male patients will become sterile, but females treated for ALL can deliver normal children. Sperm banking, which is available in major centres, is a desirable option, especially for patients undergoing bone-marrow transplantation; it should be arranged prior to commencing intensive treatment.

Overview

The acute leukemias were considered incurable 25 years ago. With optimal treatment, up to 70% of all children with ALL can now be cured. A proportion of all adults with AML (10%-25%) can expect long-term disease-free survival after conventional chemotherapy. Allogeneic bone-marrow transplantation has probably increased the proportion of patients cured who have achieved remission, and has rescued a minority with advanced disease. Recent developments in autologous marrow transplantation suggest that a greater proportion of patients may benefit from very intensive therapy. Improvements in supportive care have reduced morbidity and mortality associated with intensive regimens. However, we are still faced with the challenge of helping the remaining 60% or so of patients who die either

of their disease or from treatment-related complications. ■

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WARNINGS ASA is a frequent cause of accidental poisoning in infants and young children. ARTRIA SR (sustained-release acetylsalicylic acid tablets) should therefore be kept well out of the reach of all children.

ARTRIA SR should be administered only under medical supervision and with caution to children or teenagers with chickenpox or influenza.

PRECAUTIONS Administer salicylates cautiously to patients with asthma and other allergic conditions or those with a history of gastrointestinal ulcerations, a tendency to GI bleeding, significant anemia or hypoprothrombinemia.

Acute hepatitis has been reported rarely in patients with systemic lupus erythematosus and juvenile rheumatoid arthritis with total plasma salicylate concentrations above 25 mg/100 mL (1.8 mmol/L). Patients have recovered after cessation of therapy.

Use in Pregnancy ASA has been found to delay parturition in rats but does not appear to have any teratogenic effects. High doses of ASA (3 g daily) during pregnancy may lengthen the gestation and parturition time.

Because of possible adverse effects on the neonate and the potential increase for maternal blood loss, ASA should be avoided during the last 3 months of pregnancy, unless the potential benefits outweigh the potential risks.

Salicylates may appear in human breast milk and thus should be administered to nursing mothers with caution.

Drug Interactions ARTRIA SR should be administered with caution to patients who are taking anticoagulant medication, as ASA may potentiate its action.

Diabetics receiving concurrent salicylate and hypoglycemic therapy should be monitored closely and reduction of sulfonylurea hypoglycemic drug dosage or insulin requirements may be necessary.

At higher doses salicylate acts as a uricosuric agent. However, smaller doses of ASA may reduce uric acid clearance and decrease the uricosuric effects of probenecid, sulfipyrazone and phenylbutazone.

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Salicylates reduce the rate of methotrexate elimination by the kidney.

The risk of gastrointestinal ulceration is increased when salicylates are given concomitantly with alcohol, corticosteroids or pyrazole compounds.

Clinical Laboratory Results Salicylates may alter thyroid function tests.

ADVERSE REACTIONS Gastrointestinal: nausea, vomiting, diarrhea, gastritis, gastrointestinal bleeding and/or ulceration, dyspepsia, heartburn.

Ear: tinnitus, vertigo, hearing loss.

Hematologic: leukopenia, thrombocytopenia, purpura, anemia.

Dermatologic and Hypersensitivity: urticaria, angioedema, pruritus, skin eruptions, asthma, anaphylaxis.

Miscellaneous: acute reversible hepatotoxicity; mental confusion, drowsiness, sweating, thirst.

SYMPTOMS AND TREATMENT OF OVERDOSAGE

Symptoms Mild overdosage may include rapid and deep breathing, nausea, vomiting (leading to alkalosis), hyperpnea, vertigo, tinnitus, flushing, sweating, thirst and tachycardia. In more severe cases, acid-base disturbances, including respiratory alkalosis and metabolic acidosis, can occur. Severe cases may show fever, hemorrhage, excitement, confusion, convulsions or coma and respiratory failure.

Treatment Administer water, milk or activated charcoal and remove by cautious gastric lavage or emesis. Force fluids (e.g., salty broth) to replace sodium loss. If the patient is unable to retain fluids orally, the alkalosis can be treated by i.v. hypertonic saline. If salicylism acidosis is present, i.v. sodium bicarbonate is preferred because it increases renal excretion of salicylate. Treat severe excitement or convulsions with a barbiturate. Vitamin K is indicated if there is evidence of hemorrhage. Hypoglycemia is an occasional accompaniment of salicylate overdosage and can be managed by glucose solutions. If a hemorrhagic diathesis is evident, give vitamin K.

Hemodialysis may be useful in complex acid base disturbances, particularly in the presence of abnormal renal function. Prophylactic antibiotics are helpful.

Use general supportive measures for depressed respiration, e.g., oxygen and artificial respiration. Convulsions may best be treated by the administration of succinylcholine and artificial ventilation with oxygen. Do not use CNS depressant agents. Hyperthermia and dehydration are immediate threats to life and initial therapy must be directed to their correction and to the maintenance of adequate renal function. External cooling with cool water should be provided quickly to any child who has a rectal temperature over 40°C (104°F).

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REFERENCES

1. Comparative multiple dose bioavailability study of ASA tablets in human volunteers (study 227). Data on file.
2. Artria SR product monograph, available to physicians and pharmacists upon request.



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